

1. Overview of the Models

Model	Key Idea	Temporal Handling	Strengths	Weaknesses
LSTM	Recurrent, sequential memory	Processes sequence step by step ; retains long-term dependencies with gated memory	Can capture temporal patterns over long sequences; good for sequential data	Struggles with very long sequences, sequential bottleneck limits parallelization; gradient decay for very long dependencies
Transformer	Attention-based, non-recurrent	Processes entire sequence simultaneously ; uses self-attention to capture dependencies	Captures global context; massively parallelizable; flexible receptive field; no sequential bottleneck	Needs more data; positional encoding required to inject order; computationally heavier
TCN	Convolutional, causal temporal convolutions	Local receptive field grows via dilations	Stable, parallelizable; fixed-size kernels	Poor at very long-range dependencies; might miss global patterns in EDC decay curves

2. Nature of the Data: EDC (Energy Decay Curve)

- Long sequence representing sound energy decay in a room.
- Global correlations: early decay influences later decay.
- High dynamic range: early vs. late energy.

Implication: models need **long-range sequence awareness**.

3. Why Transformer Outperformed LSTM

a) Global context via self-attention

- Every timestep sees all others.
- Critical for metrics like **T20** and **C50** (early vs late energy).

- LSTM relies on hidden states, prone to forgetting distant timesteps.

b) Parallel computation and gradient stability

- Transformer processes sequences in parallel.
- Stable gradients → better convergence.
- LSTM is sequential → slower, harder to train on long sequences.

c) Flexible receptive field

- Attention allows any timestep to influence any other.
- Important for long sequences like 3000-point EDC.

d) Handling of linear vs log scales

- Transformer adapts naturally to varying scales.
- LSTM might overweight recent timesteps, affecting late-decay metrics.

e) Architecture suitability for regression over sequence

- EDC prediction is **sequence-to-sequence regression**.
- Transformers excel in seq2seq tasks; LSTMs need careful tuning.

4. Why TCN performed so poorly

- TCN is convolutional → local receptive field.
- Even with dilations, misses **global decay patterns**.
- Leads to huge MAE/RMSE (15000–22000), unphysical predictions.

5. Performance Comparison Table

Metric	LSTM	Transformer	Key Reason
EDT	0.020	0.021	Both capture short-term early decay equally well
T20	0.020	0.065	Transformer sees full decay; optimization may vary
C50	0.90	0.85	Transformer captures early-late energy ratio better
Training	Sequential, slower	Parallel, faster	Gradient stability + attention to full sequence
Generalization	Good	Excellent	Attention learns global room acoustic patterns

6. Key Takeaways

1. **Transformers = global attention** → best for long sequences where **early and late data must interact**.
2. **LSTMs = sequential memory** → good but limited in long, highly dynamic sequences.
3. **TCNs = local convolutions** → too local, fails for long-range dependencies.
4. **EDC prediction** is inherently a **global regression problem**, making Transformers naturally superior.